AADITYA OJHA

Civil Engineer | Engineer in Training (EIT)

562-977-7476 | medamid@gmail.com | ASCE Member | Los Angeles Metropolitan Area, CA

Civil Engineer with 5+ years of diverse, hands-on experience across transportation, geotechnical, water resources, and construction disciplines. Demonstrated ability to supervise large-scale infrastructure projects, lead multi-discipline field teams, and deliver quality-controlled outcomes in demanding international environments. Experienced with ADB-funded projects exceeding $300M, from earthworks and foundation construction to drainage systems, pipeline networks, and highway design. Currently conducting Caltrans-sponsored M.S. research (GPA: 4.0) and pursuing FE licensure.

TECHNICAL SKILLS

Design & Engineering Software: AutoCAD & Civil3D, Microstation, OpenRoads, HY-8 (FHWA), SWMM, Synchro, TransCAD, ArcGIS Pro, SAP2000, GTFS
Geotechnical & Field Tools: Soil Sampling and Classification, Field Density Testing, Compaction Control, Foundation Inspection, Total Station, GPS
Construction & Project Controls: Construction Supervision, QA/QC, Contractor Coordination, Quantity Estimation, Bills of Quantities, Project Scheduling, Materials Testing
Research & Analysis Tools: MATLAB, Minitab, LCApave, eLCAp, Machine Learning (Predictive Modeling)
Core Competencies: Pavement Design and Rehabilitation, Drainage and Stormwater Management, Water Supply Systems, Subgrade Stabilization, Highway Design, Environmental Impact Assessment, Surveying
Other: Microsoft Office Suite (Word, Excel, PowerPoint, Project), Python, C Programming

WORK EXPERIENCE

Graduate Researcher | Caltrans / California State University Long Beach *(Aug 2024 – May 2026)*

- Conducted experimental research on sustainable asphalt mixtures (RAP, crumb rubber, polymer fibers, RAS, rejuvenators), evaluating rutting, cracking, and long-term durability performance across 15+ mix designs, contributing to Caltrans pavement material standards using ASTM, AASHTO, and EPA guidelines.
- Developed predictive machine learning models to estimate pavement rutting susceptibility under varying traffic and environmental conditions, bridging data science and civil engineering practice.
- Supported Environmental Impact Assessments for state infrastructure projects, ensuring compliance with CEQA and applicable federal environmental regulations.
- Authored and presented peer-reviewed research at ASCE Airfield and Highway Pavements 2025; findings published in the ASCE conference proceedings and TRB 2026.

Civil Engineer | Eptisa Servicios de Ingenieria (Spain-based) *(Sep 2021 – Aug 2024)*

- Supervised $300M+ ADB-funded infrastructure construction encompassing highway pavements, underground storage tanks and chambers, drainage networks, and water supply pipelines, coordinating across structural, geotechnical, and civil disciplines simultaneously.
- Led a field team of 15+ Engineers in Training and 4 Civil Engineers; directed daily site operations, enforced QA/QC protocols, and ensured compliance with international design and safety standards with AutoCAD for pipeline and drainage design.
- Oversaw earthworks and foundation preparation including subgrade stabilization, compaction control, excavation shoring, waterproofing, and backfilling for underground structures.
- Conducted precision surveying with Total Stations and GPS for progress verification, as-built documentation, and quantity assessments; executed quality testing for structural concrete and water pipeline systems.

Civil Engineer | DOHWA Engineering Pvt. Ltd. (Korea-based) *(Nov 2019 – Sep 2021)*

- Managed installation of a 1,056-mile underground pipeline network (Melamchi Water Supply Project), overseeing excavation, bedding, pipe laying, joint inspection, compaction, and backfilling beneath urban road corridors.
- Supervised concrete placement operations including vibration, curing, and finishing; collected and tested field specimens for compressive strength (7-day and 28-day), documenting results and initiating corrective action on non-conforming batches.
- Directed rehabilitation of 60+ miles of rigid pavements, coordinating subbase preparation, concrete placement, and surface restoration while managing crews and equipment across multiple work fronts supporting design and as-built drawings in Microstation.
- Led design and construction of overhead and underground water storage tanks; coordinated geotechnical investigations, reviewed soil reports, and optimized foundation systems to meet structural performance requirements.

Surveying Engineer | ENGIC Group of Company (Nepal) *(Jan 2019 – Oct 2019)*

- Executed topographic and alignment surveys using Total Stations and GPS for roadway and structural design projects, capturing horizontal and vertical control data for engineering design use.
- Processed field data into topographic models supporting earthwork calculations, route alignment design, and construction planning for infrastructure development projects.

EDUCATION

Master of Science, Civil Engineering | California State University, Long Beach | GPA: 4.0/4.0 *(Aug 2024 – May 2026)*
Bachelor of Engineering, Civil Engineering | Tribhuvan University, Nepal *(Sep 2015 – Oct 2019)*

LICENSES & CERTIFICATIONS

ASCE: Member, American Society of Civil Engineers (Since January 2025)
FE Exam: Fundamentals of Engineering, Civil Discipline | Scheduled: April 2026, California

PUBLICATIONS

Ojha, A. et al. (2025). Employing LCA in Asphalt Mixture Design Using RAP. ASCE, Airfield and Highway Pavements 2025.
https://doi.org/10.1061/9780784486221.006

RELEVANT COURSEWORK

Construction Management | Foundation Engineering | Advanced Soil Mechanics | Pavement Engineering | Traffic Engineering | Transportation Planning and Engineering | Water Supply Engineering | Drainage Design | Stormwater Management | Environmental Impact Assessment | Design of RCC and Steel Structures | Estimating and Costing | AI/ML in Engineering | Engineering Economics